



Building Generator

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Creating Buildings

Tool Usage:

This tool is used to create buildings from a modular mesh kit straight into a level into UE.

Setup

Drag and drop the HDA into Unreal and then into the level. Then adjust the mode:



Blockout Mode



Reference Mode

Core Settings:

Under General you can find everything you need to set the building size

Applying Randomisation:

This tool generates 8 randomised float values that can be used to drive vertex painting, masking and colour. Here's a comparison of the result on brickwork and windows.



Without Randomisation



With Randomisation

Working with Grammar

What is Grammar?

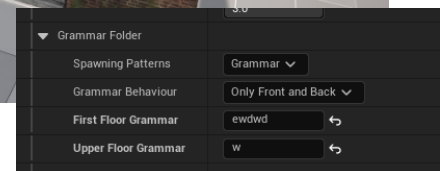
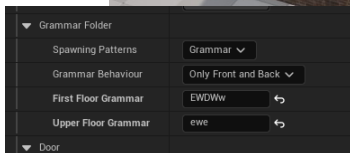
Grammar allows you to define a wall piece based on its reference, use the following:

E = Empty

D = Door

W = Window

G = Gutter or D for downpipe (**pipes category only**)



Adjusting Grammar Behaviour

Grammar loops on point direction, this can be reversed or randomised so you do not need to define specific grammar. Slight adjustments can create variations quickly.

You can reverse the grammar, mirror or create broken spawns



Broken Variant

You can toggle spawning all parts as broken, half a building or randomly picked pieces.

Additional Generators

Constructing a Clothesline

To build drag out the tool and add a curve input. Drag each point on the curve to the desired start and end and begin to randomise generated values.



Constructing Telegram Poles

Drag out the tool and add some cubes to the level.

These cubes will define the attachment points and can be deleted later after baking.

Then add a world input and assign the cubes to the selection. You can define how many cables are created from the attached meshes as well.

To connect different poles together, spawn an attachment point at the top and bake that out of the actor, then on still live tools assign that to the world selection.

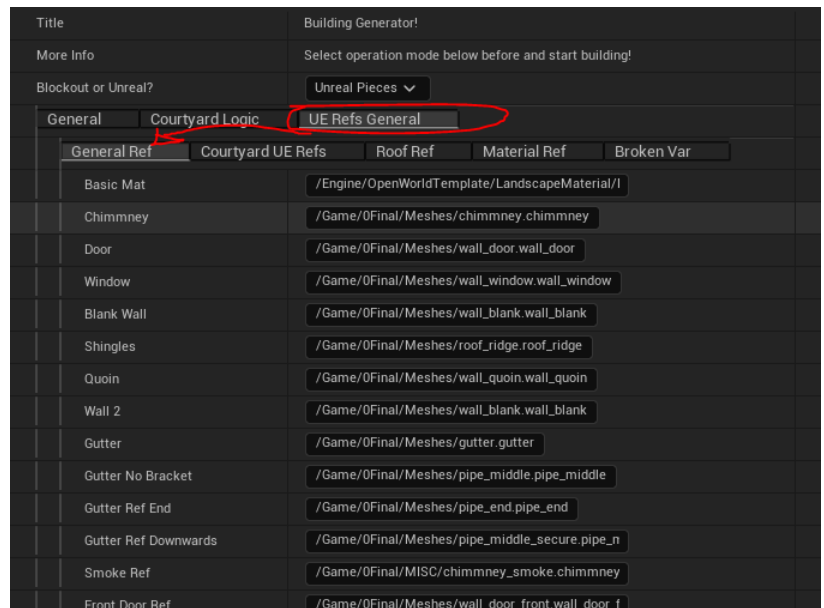


UE References and Optimisation

Swapping References

To customise the tool to work with your kit, ensure you replace the required references in the unreal references tab.

You also need to adjust the modular dimensions found in the general tab to fit the new kit. Your kit needs to remain consistent in width and height for the best results when placing static meshes and instancing.



Further Optimisation Considerations:

To further optimise the tool for your need consider editing the tool HDA in Houdini to reduce the number of custom floats.

Generate a simplified HLODS and enable lods on your modular pieces that replace high cost shaders such as windows with opaque variants at a distance.

Also consider using packed level actors for a more cheaper way to add costly buildings into your level as a still split up actor if necessary.

